

Lecture 1

Introduction

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Definition 1: A set $\mathbb{F}$ is called a field if $\mathbb{F}$ has:

1. A zero element (denoted 0).
2. A one element (denoted 1).
3. A defined notion of addition on elements of the set.
4. A defined notion of multiplication on elements of the set.

The addition and multiplication on $\mathbb{F}$ must satisfy, for all $a, b, c \in \mathbb{F}$:

1. Commutativity.
 - $a + b = b + a$
 - $ab = ba$.
2. Associativity.
 - $(a + b) + c = a + (b + c)$
 - $(ab)c = a(bc)$.
3. Distributivity over addition.
 - $a(b + c) = ab + ac$.

As well, addition and multiplication on $\mathbb{F}$ must:

1. Have an identity element (0 and 1 respectively) such that for all $a \in \mathbb{F}$.
 - $a + 0 = 0 + a = a$
 - $1 \cdot a = a \cdot 1 = a$.
2. Have an inverse element.
 - $\forall a \in \mathbb{F}, \exists b \in \mathbb{F}$ such that $a + b = 0$
 - $\forall a \in \mathbb{F} \setminus \{0\}, \exists b \in \mathbb{F}$ such that $ab = 1$

Example 1: Is the set of all natural numbers ($\mathbb{N}$) a field?

Solution:

No! There does not exist an element $a \in \mathbb{N}$ such that $1 + a = 0$. In other words, no elements besides 0 in $\mathbb{N}$ have an additive inverse.

Example 2: Is the set of all integers ($\mathbb{Z}$) a field?

Solution:

No! There does not exist an element $a \in \mathbb{Z}$ such that $2a = 1$. In other words, no elements besides 1 in $\mathbb{Z}$ have a multiplicative inverse.

Example 3: Is the set of all rational numbers ($\mathbb{Q}$) a field?

Solution:

Yes!

Example 4: Is the set of all real numbers ($\mathbb{R}$) a field?

Solution:

Yes! Although the proof is quite complicated. In many undergraduate courses, $\mathbb{R}$ is often defined axiomatically as a complete, ordered field.

Example 5: Is the set of all complex numbers ($\mathbb{C}$) a field?

Solution:

Yes!